HDL-MR0410.431

4CH 10A High Power Switch Actuator

buspro

**Datasheet**

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Figure 1. 4CH 10A High Power Switch Actuator

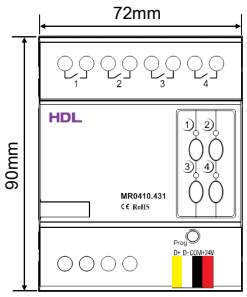



Figure 2. Dimensions - Front View

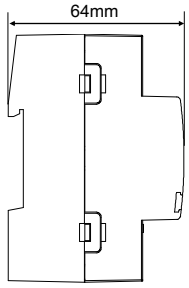



Figure 3. Dimensions - Side View

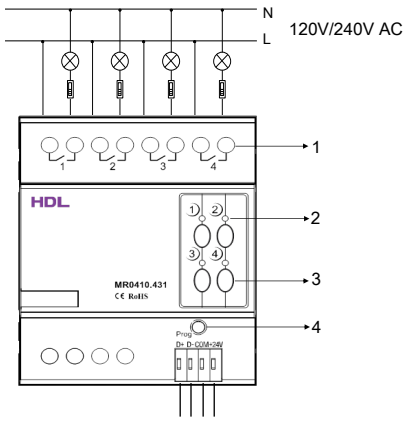



Figure 4. Wiring

## Overview

4CH 10A High Power Switch Actuator (See Figure 1) has 4 channels (10A for each channel) and 16A magnetic latching relay. With low power consumption and long lifetime, the module has manual control switches and supports scene and sequence control.

## Functions

- 4-channel relay output unit
- Scene control and sequence control
- Up to 4 separate zones.
- 8 scenes for each zone, scene running time up to 60 minutes
- 2 sequences for each zone, up to 4 steps for each sequence
- Light protection delay for each channel (0-60 minutes)
- 0-25s of bulk power-on time for each channel.
- Manual switch for each channel
- Selected scene or scene before power off automatically activated when device restarts
- LED indicator: indicates current channel status.
- HDL Buspro communication
- Supports online upgrade

## Important Notes

- Buspro cable - CAT5E or dedicated HDL Buspro cable
- Buspro connection - Series connection (hand-in-hand recommended).
- Maximum current of each output channel: 10A
- One 10A breaker for each channel for protection

## Product Information

**Dimensions - See Figure 2 - 3**

**Wiring - See Figure 4**

1. Connect to load
2. LED indicator indicates the channel status
3. Direct connection switch
4. Programming button & module indicator

Indicator flashes when device is in normal working mode. Press the button for 3 seconds, then the ID can be read and modified via HDL Buspro Setup Tool.

**Installation - See Figure 5 - 7**

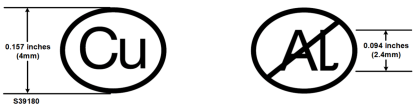
Step 1. Fix the DIN rail with screws.

Step 2. Buckle the bottom cap of the actuator on the edge of the DIN rail.

Step 3. Press the device on the DIN rail, slide it and fix it up until an appropriate position is adjusted.

## Safety Precautions

- The installation and testing for the product must be carried out by HDL Automation Co., ltd. or its appointed service agencies. The electric construction shall comply with local laws and safety regulations.
- The device should be installed with DIN rail in DB box. HDL will not be responsible for any consequence caused by the inexpert or faulty installation and wiring methods, which are not in accordance with the instructions contained in this operating instruction.
- Please do not privately disassemble or replace any parts of the product. Otherwise, it may cause mechanical fault, electric shock, fire or personal injuries.
- Please contact our after-sales departments or our designated service agencies for your maintenance service. Product failures caused by private disassembly are not subject to this warranty.
- It is not allowed to exceed the range.
- CAUTION - Risk of Electric Shock - More than one disconnect switch may be required to de-energize the equipment before servicing.
- The marking appears on the device, shown below shall be used to indicate that the device is for use with copper wire. The marking shall be legible with letters at least 2.4 mm high. “Use copper wire only”, “Cu wire only” or equivalent wording, or a marking containing both the symbols as the illustrations.




## Package Contents

HDL-MR0410.431*1 / Buspro connector*1 / Label*5 / Datasheet*1



Figure 5

1

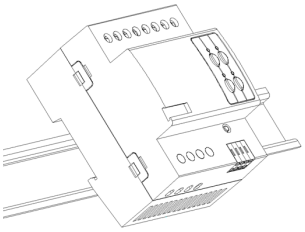


Figure 6

2

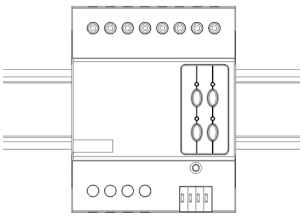


Figure 7

Figure 5 – 7. Installation

# Technical Data

| Basic Parameters | |
|---|---|
| Working voltage | 15~30V DC Class 2 |
| Working current | 40mA/24V DC |
| Input voltage | 120V/240V AC (50/60Hz) |
| Output channel | 4CH/10A |
| Relay | 16A Magnetic latching relay |
| Maximum current of each channel | 10A |
| Electronic life time of relay | >60000 (Resistance load) |
| Protection | Connect a breaker in each channel |
| **External Environment** | |
| Working temperature | -5℃~45℃ |
| Working relative humidity | ≤90% |
| Storage temperature | -20℃~60℃ |
| Storage relative humidity | ≤93% |
| **Specifications** | |
| Dimensions | 72mm×90mm×64mm |
| Net weight | 252g |
| Housing material | Nylon, PC |
| Installation | 35mm DIN rail installation<br>(See Figure 5 - 7) |
| Protection rating (Compliant with EN 60529) | IP20 |

| Recommended Load Types and Power |
|---|
| 240V, 10A, Resistive, 100,000 cycles, 40°C; |
| 240V, 1HP (8FLA/48LRA), Motor, 6,000 cycles, 40°C; |
| 240V, 6A, Standard Ballast, 6,000 cycles, 40°C; |
| 120V, 0.5HP (9.8FLA/58.8LRA), Motor, 20,000 cycles, 40°C; |
| 120V, 10A, Electronic Ballast, 20,000 cycles, 40°C; |
| 120V, 10A, Standard Ballast, 6,000 cycles, 40°C; |

**Name and Content of Hazardous Substances in Products**

| Components | Hazardous substances | | | | | |
|---|---|---|---|---|---|---|
| | Lead (Pb) | Mercury (Hg) | Cadmium (Cd) | Chromium VI (Cr (VI)) | Poly-brominated biphenyls (PBB) | Poly-brominated diphenyl ethers ( PBDE ) |
| Plastic | o | o | o | o | o | o |
| Hardware | o | o | o | o | - | - |
| Screw | o | o | o | × | - | - |
| Solder | × | o | o | o | - | - |
| PCB | × | o | o | o | o | o |
| IC | o | o | o | o | × | × |

The symbol “-” indicates that the hazardous substance is not contained.

The symbol "o" indicates that the content of the hazardous substances in all the homogeneous materials of the component is below the limit requirement specified in the Standard IEC62321-2015.

The symbol “×” indicates that the content of the hazardous substance in at least one of the homogeneous materials of the part exceeds the limit requirement specified in the Standard IEC62321-2015.

# HDL Buspro Cable Guide

| HDL Buspro | HDL Buspro Cable | CAT5/CAT5E |
|---|---|---|
| DATA+ | Yellow | Blue/Green |
| DATA- | White | Blue white/Green white |
| COM | Black | Brown white/Orange white |
| 24V DC | Red | Brown/Orange |

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